

Business Problem Statement & Requirements Specification

Comprehensive analytical framework for workforce planning, attrition analysis, demographic intelligence, and employee satisfaction tracking.

DOMAIN

Human Resources / People Analytics

DOCUMENT VERSION

v1.0 (Production Blueprint)

SCOPE

Executive KPIs & Interactive Dashboard Visuals

1. Context & Business Problem Overview

The Human Resources (HR) department plays a critical operational and strategic role in monitoring and managing employee data to maintain a healthy, productive, and resilient workforce. Currently, the organization suffers from a **critical lack of standardized metrics, visibility, and business intelligence capabilities** across its workforce lifecycle.

Without a centralized reporting engine, leadership cannot reliably determine the organization's true headcount, differentiate active staff from exited personnel, or quantify attrition dynamics. This absence of clear Key Performance Indicators (KPIs) and exploratory visualization tools impedes workforce planning, obscures retention vulnerabilities across demographics and departments, and prevents proactive talent retention interventions.

2. Key Performance Indicators (KPIs) Requirement

To resolve the baseline visibility gap, the business requires five foundational headline KPI cards to be calculated and monitored continuously:

KPI Indicator	Identified Business Problem & Gap	Operational Impact & Objective
1. Employee Count	The HR department lacks visibility into the total number of employees, making it challenging to assess workforce size and plan for future growth or downsizing effectively.	Establishes the authoritative workforce baseline needed for organizational capacity planning and headcount forecasting.
2. Attrition Count	The organization lacks a standardized method to track employee attrition, resulting in incomplete and unreliable data on the number of employees who have left the organization.	Delivers an audited count of exits across defined time horizons to eliminate data discrepancies and enable accurate turnover tracking.
3. Attrition Rate	Without a clear measure of attrition rate, the organization cannot assess the overall turnover level or compare it with industry benchmarks, hindering the ability to gauge employee satisfaction and engagement.	Normalizes departures against total headcount ((Attrition Count / Total Employees)

KPI Indicator	Identified Business Problem & Gap	Operational Impact & Objective
		× 100) to evaluate turnover severity against industry norms.
4. Active Employees	The organization lacks a mechanism to differentiate between active and inactive employees, leading to difficulties in accurately assessing the current workforce's productivity and capacity.	Provides real-time operational capacity metrics by isolating active payroll contributors from on-leave, suspended, or archived records.
5. Average Age	The HR department lacks visibility into the average age of employees, making it difficult to evaluate workforce demographics, succession planning, and the organization's ability to attract and retain younger talent.	Guides long-term workforce sustainability, leadership succession pipelines, and generational recruiting/retention programs.

3. Visual Analytics & Reporting Requirements (Chart Specifications)

In addition to headline metrics, leadership requires multi-dimensional visual analyses to diagnose root causes of turnover and demographic shifts:

1. Attrition by Gender

Donut / Split Bar Chart

Problem & Need: The HR department faces challenges in understanding attrition patterns based on gender, making it difficult to identify gender-related disparities and implement targeted retention strategies. Visualizing gender-stratified attrition will uncover potential DEI (Diversity, Equity, and Inclusion) friction points.

2. Department-wise Attrition

Horizontal Bar Chart

Problem & Need: The HR department lacks visualizations to showcase attrition rates across different departments. This hinders their ability to identify departments with higher attrition rates and address underlying structural, managerial, or operational concerns effectively.

3. Number of Employees by Age Group

Column Histogram / Distribution Chart

Problem & Need: The HR department requires visual representations to analyze the distribution of employees across various age brackets (e.g., Under 25, 25–34, 35–44, 45–54, 55+). This helps assess workforce demographics, highlight age gaps or retirement risks, and design targeted talent policies.

4. Job Satisfaction Ratings

Heatmap / Matrix / Segmented Bar

Problem & Need: The HR department lacks visualizations to represent job satisfaction ratings (spanning job roles, compensation bands, and work environments), hindering their ability to measure employee sentiment, engagement levels, and turnover correlations effectively.

5. Education Field-wise Attrition

Bar / Tree-Map Chart

Problem & Need: Visual representations are needed to analyze attrition rates by employee educational backgrounds (e.g., Technical, Business, Medical, Humanities). This identifies specific degree cohorts associated with higher exit velocities and enables tailored compensation or career-pathing plans.

6. Attrition Rate by Gender for Different Age Groups

Clustered Column / Multi-series Line Chart

Problem & Need: The department lacks cross-tabulated visualizations displaying attrition rates segmented simultaneously by gender and age cohort. This granular intersectional view is vital for pinpointing vulnerable talent segments (e.g., early-career female employees or mid-career technical leads).

4. Strategic Business Impact & Intended Outcomes

Proactive Talent Retention

Transform HR from reactive firefighting to predictive intervention by isolating departments, roles, and demographics suffering elevated exit rates.

Workforce Planning & Succession

Accurate active employee counts and age distributions ensure seamless succession pipelines and prevent sudden institutional knowledge drains.

Benchmarking & Performance Audits

Enable leadership to contrast internal attrition metrics against market and industry baselines to identify systemic cultural or compensation shortcomings.

Evidence-Based HR Policies

Ground diversity, compensation, training, and wellness initiatives in validated empirical data rather than qualitative assumptions.